

Bilkent University EEE202 LAB #3

Passive and Active Band-Pass Filters

Preliminary Work

Passive Low-Pass Filter Design

Design a third-order (maximally flat) Butterworth low-pass filter using two capacitors and one inductor, as suggested in the figure below.

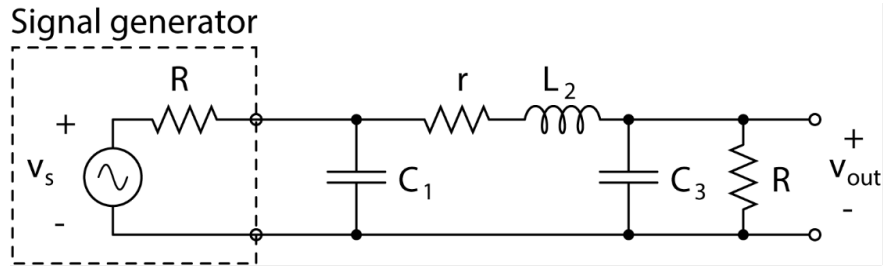


Figure 1: Low-pass filter circuit representation.

In this lab, you will be assigned a specific inductor chosen from the nominal inductance (L_{nom}) values listed in Tables 1 and 2. Unshielded and shielded tunable inductors are available in the lab. Shielded inductors prevent magnetic coupling between neighboring inductors since they have a ferromagnetic cover. The inductors are tunable using a tuning stick. The Q values of the inductors are shown in the Tables 1 and 2. The series resistance of the inductor can be determined by

$$r = \frac{2\pi f_T L_{\text{nom}}}{Q}, \quad (1)$$

where f_T is the test frequency.

Inductors will be distributed during your scheduled lab session (upon signing for them) and must be returned at the end of that session. **Please note that these inductors are only available during your assigned lab time; you will not be able to obtain them beforehand or afterward.**

Part Number	Color	$L_{\min}$ (μH)	$L_{\max}$ (μH)	L_{nom} (μH)	Min Q @ L_{nom}	Test Frequency
SLOT TEN-2-01	Brown	0.80	1.7	1.2	47	7.9 MHz
SLOT TEN-2-02	Red	1.25	2.75	2.0	48	7.9 MHz
SLOT TEN-2-03	Orange	1.75	4.0	2.9	48	7.9 MHz
SLOT TEN-2-04	Yellow	2.4	5.4	3.9	47	7.9 MHz
SLOT TEN-2-05	Green	3.5	7.8	5.6	47	7.9 MHz
SLOT TEN-2-06	Blue	4.7	10.6	7.6	46	7.9 MHz
SLOT TEN-2-07	Violet	7.4	15.6	11.5	38	2.5 MHz
SLOT TEN-2-08	Grey	11	25	18.2	40	2.5 MHz
SLOT TEN-2-09	White	16	35	25.3	40	2.5 MHz
SLOT TEN-2-10	Black	22	50	36.5	40	2.5 MHz
SLOT TEN-2-11	Brown	33	72	52.5	39	2.5 MHz
SLOT TEN-2-12	Red	46	103	74.8	51	790 KHz
SLOT TEN-2-13	Orange	66	136	100	40	790 KHz
SLOT TEN-2-14	Yellow	95	198	146	44	790 KHz
SLOT TEN-2-15	Green	136	297	216	40	790 KHz
SLOT TEN-2-16	Blue	198	426	312	45	790 KHz
SLOT TEN-2-17	Violet	286	630	530	33	790 KHz
SLOT TEN-2-18	Grey*	418	927	790	38	790 KHz

Table 1: Unshielded With Ferrite Core

Part Number	Color	$L_{\min}$ (μH)	$L_{\max}$ (μH)	L_{nom} (μH)	Min Q @ L_{nom}	Test Frequency
SLOT TEN-3-01	Brown	0.70	0.94	0.82	65	7.9 MHz
SLOT TEN-3-02	Red	1.05	1.41	1.23	37	7.9 MHz
SLOT TEN-3-03	Orange	1.5	2.0	1.75	40	7.9 MHz
SLOT TEN-3-04	Yellow	2.04	2.76	2.4	40	7.9 MHz
SLOT TEN-3-05	Green	2.9	3.9	3.4	39	7.9 MHz
SLOT TEN-3-06	Blue	4.25	5.75	5.0	38	7.9 MHz
SLOT TEN-3-07	Violet	6	8.2	7.1	35	7.9 MHz
SLOT TEN-3-08	Grey	9.4	12.6	11.0	25	2.5 MHz
SLOT TEN-3-09	White	12.8	17.2	15.0	25	2.5 MHz
SLOT TEN-3-10	Black*	18.7	25.3	22.0	26	2.5 MHz
SLOT TEN-3-11	Brown*	26.4	35.6	31.0	25	2.5 MHz
SLOT TEN-3-12	Red*	37.4	50.6	44	29	2.5 MHz
SLOT TEN-3-13	Orange*	52.7	71.3	62	30	2.5 MHz
SLOT TEN-3-14	Yellow*	79	105	92	28	2.5 MHz
SLOT TEN-3-15	Green*	108.8	147.2	128	18	790 KHz
SLOT TEN-3-16	Blue*	155	208	182	20	790 KHz
SLOT TEN-3-17	Violet*	230	310	270	16	790 KHz

Table 2: Shielded With Ferrite Core

1. Determine the capacitor values and the 3-dB cutoff frequency (f_0) of the filter using the formulas

$$f_0 = \frac{b_2 R}{2\pi L_2}, \quad (2)$$

$$C_1 = \frac{b_1}{2\pi f_0 R}, \quad C_3 = \frac{b_3}{2\pi f_0 R}. \quad (3)$$

where b_1, b_2 , and b_3 are the Butterworth filter prototype element values and L_2 is the assigned inductance value. For a third-order filter, $b_1 = 1$, $b_2 = 2$, and $b_3 = 1$ as shown in the Table 3 below. The load and source resistances are both $R = 50\Omega$.

2. Plot the theoretically expected transfer function for an $n = 3$ Butterworth filter using MATLAB. The transfer function in dB is given by

$$\left| \frac{v_{\text{out}}}{v_s} \right|_{\text{dB}} = -10 \log_{10} \left(1 + \left(\frac{f}{f_0} \right)^{2n} \right). \quad (4)$$

n	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
1	2.000							
2	1.4142	1.4142						
3	1.0000	2.0000	1.0000					
4	0.7654	1.8478	1.8478	0.7654				
5	0.6180	1.6180	2.0000	1.6180	0.6180			
6	0.5176	1.4142	1.9319	1.9319	1.4142	0.5176		
7	0.4450	1.2470	1.8010	1.8010	1.2470	0.4450		
8	0.3902	1.1111	1.6629	1.9616	1.6629	1.1111	0.3902	

Table 3: Butterworth prototype coefficients for orders $n = 1$ to 8.

3. Perform an LTSpice AC analysis of the designed filter (including the r -value of the chosen inductor) to determine the filter's transfer function, v_{out}/v_s , in dB as a function of frequency. In LTSpice, set the AC amplitude of v_s to 2 (so that the passband gain is unity). Determine the 3-dB cutoff frequency from the simulated response and calculate the filter's insertion loss (in dB) at $f_0/10$ (due to the non-zero r value). Also, verify that the filter exhibits a -18 dB/octave roll-off in the stopband.

Active Band-Pass Filter Design

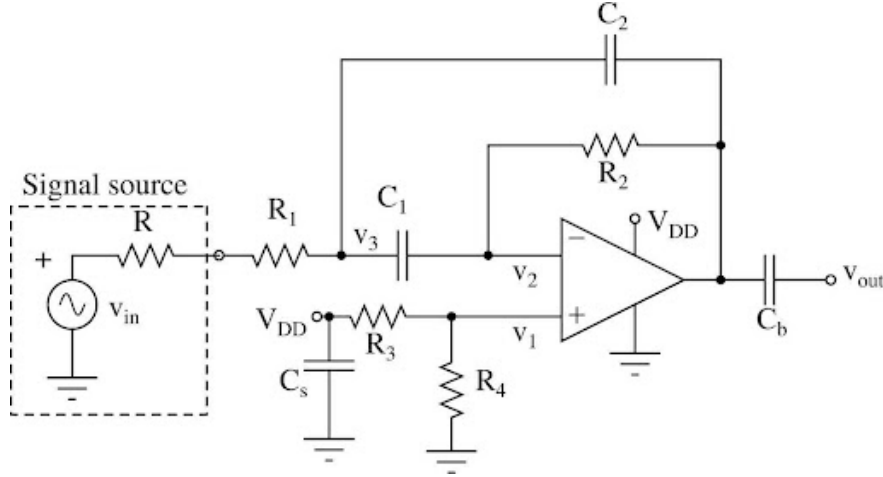


Figure 2: Active band-pass filter circuit design.

- Design an active band-pass filter with a center frequency f_0 and a bandwidth BW . You may use the circuit shown in the Figure 2, which is suitable for low frequencies with narrow bandwidths. The OPAMP in this circuit uses a single supply voltage. To ensure the input and output voltages remain within 0 to V_{DD} , the positive input is biased at $V_{DD}/2$ by choosing $R_3 = R_4$ (with values greater than 1 k Ω). The output and the negative input pins are kept at the same DC level by resistor R_2 (all centered at $V_{DD}/2$). A supply bypass capacitor C_s and a DC block capacitor C_b are also included. With $C_1 = C_2$ and $R'_1 = R_1 + R$, the center frequency and Q factor are given by

$$f_0 = \frac{1}{2\pi C_1 \sqrt{R'_1 R_2}}, \quad Q = \frac{f_0}{BW} \approx \sqrt{\frac{R_2}{4R'_1}}, \quad (5)$$

and at resonance the voltage gain is

$$\text{Maximum Gain: } |A_v| = \max_{\omega} \left| \frac{v_{out}}{v_{in}} \right| = \frac{R_2}{2R'_1} \quad (6)$$

$$\text{where } A_v = -\frac{R_2}{2R'_1} \quad \text{due to inverting OPAMP} \quad (7)$$

Note that this circuit is not suitable for high-frequency applications due to the gain-bandwidth limitations of the OPAMP. For these formulas to be valid, the condition $4f_0Q^2 < (\text{gain-bandwidth product of the OPAMP})$ must be met.

- Prove the approximate formulas for the maximum gain and the center frequency (no need for Q-factor) by deriving the transfer function v_{out}/v_{in} in the frequency domain for small signals. (Hint: Write the node equations for nodes v_2 and v_3 to express v_{out} in terms of v_{in} . Assume that OPAMP is not saturated and that $v_1 = v_2 = 0$)

6. The required center frequency and bandwidth are determined by the modulus 20 of your ID number. The values are as follows:

mod(IDNo,20)	f_0 (kHz)	BW (kHz)	mod(IDNo,20)	f_0 (kHz)	BW (kHz)
0	1.0	0.10	10	3.0	0.52
1	1.2	0.13	11	3.2	0.57
2	1.4	0.17	12	3.4	0.63
3	1.6	0.20	13	3.6	0.68
4	1.8	0.24	14	3.8	0.74
5	2.0	0.28	15	4.0	0.80
6	2.2	0.33	16	4.2	0.86
7	2.4	0.37	17	4.4	0.92
8	2.6	0.42	18	4.6	0.98
9	2.8	0.47	19	4.8	1.05

Table 4: Active band-pass filter center frequency and bandwidth versus mod(IDNo,20).

Select component values and perform an LTSpice simulation to verify your design. Use the `opamp2` symbol and LM358 as the OPAMP. Include the file `LM358.txt` in LTSpice using the directive

`.include LM358.txt`

Choose $R_1 > 500\Omega$ and standard values for resistors and capacitors (e.g. multipliers, 1, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2). Note that $R = 50\Omega$ is the source resistance, and $R'_1 = R_1 + 50$. Perform an AC analysis spanning from $0.1f_0$ to $10f_0$ to verify the center frequency, bandwidth (within an error of $\pm 10\%$), and gain at f_0 . You may have to tweak the values of capacitors and/or resistors to reach the goals since the OPAMP is not ideal. Find the gain at the center frequency and compare it with the expected value.

Also, verify that the filter exhibits an asymptotic response of $+6\text{dB/octave}$ at low frequencies and -6dB/octave at high frequencies.

LM358 has a gain-bandwidth product of about 1MHz. This means the maximum gain it can generate is 10 at 100KHz. Hence, we cannot use this OPAMP with a bandpass filter for center frequencies larger than $\approx 500\text{KHz}$.

Download LM358.txt from Moodle page and place the file in the active directory.

Experimental Work

Passive Low-Pass Filter

1. Obtain the chosen inductor from the lab TA. Measure its value (L) and quality factor (Q) using an LCR meter at 100 kHz. Calculate the r value from Equation 1 by using measured L , Q and set $f_T = 100\text{kHz}$.
2. Set up your filter circuit. Use $R_0 = 47\Omega$ as the output load resistance.
3. Measure the transfer function $\frac{v_{\text{out}}}{v_{\text{in}}}$ where v_{out} is the peak voltage across the output resistance and v_{in} is the value shown on the signal generator. Use the EXT SYNC facility on the oscilloscope.
4. Determine the 3-dB cutoff frequency and choose f_0 as that value in the table below.

f_0 (kHz)	
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Passive Filter Measurement Table:

f_0	Freq (kHz)	v_{out} (V)	$v_{\text{out}}/v_{\text{in}}$	$v_{\text{out}}/v_{\text{in}}$ (dB)
0.1				
0.2				
0.4				
0.8				
1.0				
1.2				
1.4				
1.6				
1.8				
2.0				
4.0				
6.0				
8.0				
10.0				

Table 5: Measured response of the passive low-pass filter.

Note: The values 0.1, 0.2, 0.4, etc., correspond to fractions of the center frequency f_0 (i.e., $0.1 \times f_0$, $0.2 \times f_0$, etc.).

5. Determine the insertion loss (in dB) of your filter at $0.1f_0$:

Freq (kHz)	Insertion Loss (dB)
$0.1f_0$	

Table 6: Insertion loss measurement at $0.1f_0$.

6. Verify that the filter exhibits a -18 dB/octave roll-off in the reject region. Plot the dB gain on a semilog frequency axis (using MATLAB) and compare it with the LTSpice simulation.

Return the inductor to the lab technician or TA after completing your measurements.

Active Band-Pass Filter

1. Build the band-pass filter circuit on the breadboard using your designed component values. Refer to the LM358 in Figure 3.

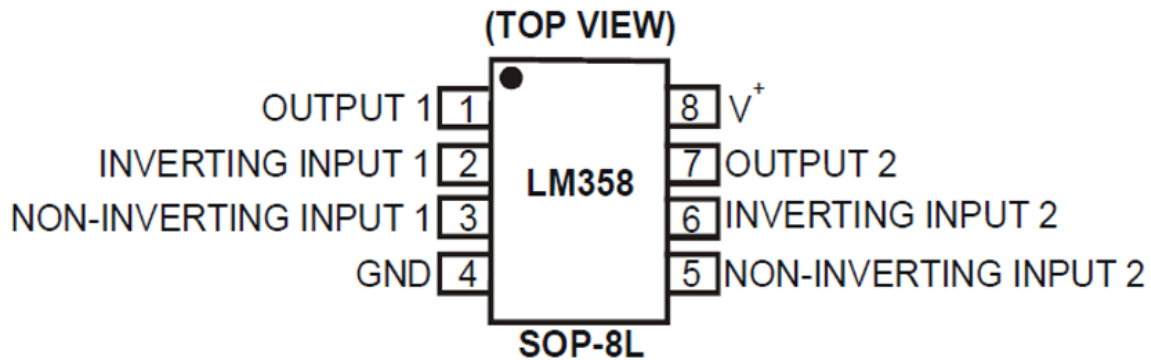


Figure 3: LM358 pinout

2. Measure the transfer function $\frac{v_{out}}{v_{in}}$ where v_{out} is the peak-to-peak voltage across the output load and v_{in} is twice the peak-to-peak value shown on the signal generator. Use EXT SYNC on the oscilloscope. (If the filter gain is high, reduce the input voltage to avoid saturating the OPAMP.) For example, if the voltage gain is 200, you should adjust the input voltage to 40 mV pp. If the voltage gain is 50, the input voltage can be as high as 200 mV pp.

- Determine the center frequency (where gain is maximum) and choose f_0 as that value in the table below.

f_0 (kHz)	
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Active Filter Measurement Table:

f_0	Freq (kHz)	v_{out} (V)	$v_{\text{out}}/v_{\text{in}}$	$v_{\text{out}}/v_{\text{in}}$ (dB)
0.10				
0.20				
0.40				
0.80				
0.90				
0.95				
1.00				
1.05				
1.10				
1.20				
1.80				
2.50				
5.00				
10.0				

Table 7: Measured response of the active low-pass filter.

Note: The values 0.1, 0.2, 0.4, etc., correspond to fractions of the center frequency f_0 (i.e., $0.1 \times f_0$, $0.2 \times f_0$, etc.).

- Determine the 3-dB bandwidth and compare it with the intended value. Also, compare the measured gain at the center frequency with the expected value.
- Verify that the filter exhibits asymptotic responses of +6 dB/octave at low frequencies and -6 dB/octave at high frequencies.
- Plot the measured dB gain on a semilog frequency axis (using MATLAB) and compare with LTSpice results.

Note on experimental issues:

- Please prefer **polyester capacitors** over **ceramic capacitors**.
- The most common issues are related to **connections**, especially when using a **bread-board**. Therefore, **soldering is strongly advised**. Ensure that **all connections are secure**. At least you can tie them in midair. Sample implementation